

1. An insurance company wants to determine the premium amount for customers based on factors like vehicle type, driving history, and past claims.

- a) What machine learning technique should be used?
- b) List the steps involved in building the model.

Model type - Regression

Step 1: Collect the insurance dataset using `read_csv` function which includes vehicle type, driving history & past claims

Step 2: Data processing - handle missing data using `imputer` & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build Regression model using linear regression, SVR, decision tree

Step 6: evaluate the model accuracy using `RSquare`, `SSE` or `SSR` to figure out which model is best to predict the premium amount to be paid

Step 7: Deploy the trained best model to predict the premium amount

2. A bank wants to predict whether a loan applicant is likely to default based on their income, credit history, and previous loans.

- a) What type of machine learning should be used?
- b) Describe the steps to train the model.

Model type: Classification

Step 1: Collect the loan dataset using `read_csv` function which includes income, credit history and previous loans

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: evaluate the model using accuracy, `f1`, classification matrix, precision to figure which model is best

Step 7: Deploy the best model to predict whether the user is default or not

3. A retail store wants to group customers based on purchasing behavior to target them with personalized offers.

- a) Which ML approach is appropriate?
- b) Explain the step-by-step process.

Model type - Clustering

Step 1: Collect the customer dataset using read_csv function

Step 2: Data processing - handle missing data & encode categorical data

Step 3: Create a model using clustering methods like DBSCAN, OPTICS, agglomerative clustering, K means etc.

Step 4: Add Hypertuning to the model & using elbow method choose the best model

Step 5: Deploy the best model & plan for the personalised offers

4. A smart energy company wants to forecast electricity consumption for households based on historical usage and weather conditions.

- a) Identify the machine learning technique to use.
- b) Outline the steps to develop the model.

Model type - Regression

Step 1: Collect the electricity dataset using read_csv function which includes historical usage, & weather conditions

Step 2: Data processing - handle missing data using imputer & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build Regression model using linear regression,SVR, decision tree

Step 6: evaluate the model accuracy using RSquare, SSE or SSR to figure out which model is best

Step 7: Deploy the best model to predict the future electricity consumption

5. An e-commerce company wants to detect fraudulent transactions using historical payment data.

- a) What ML approach is needed?
- b) Describe the steps involved.

Model type: Classification

Step 1: Collect the transactions dataset using `read_csv` function which includes income, credit history and previous loans

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: evaluate the model using accuracy, f1, classification matrix, precision to figure which model is best

Step 7: Deploy the best model to predict whether the transaction is fraudulent or not

6. A streaming platform wants to group movies into categories based on viewer ratings and watch history.

- a) What machine learning technique is best?
- b) Explain the process to implement it.

Model type - Clustering

Step 1: Collect the movies dataset using `read_csv` function

Step 2: Data processing - handle missing data & encode categorical data

Step 3: Create a model using clustering methods like DBSCAN, OPTICS, agglomerative clustering, K means etc.

Step 4: Add Hypertuning to the model & using elbow method choose the best model

Step 5: Deploy the best model & start categorizing movies

7. An online store wants to forecast its future sales revenue based on past sales trends and seasonal variations.

- a) What ML technique should be used?
- b) Detail the steps to build the model.

Model type - Regression

Step 1: Collect the sales dataset using read_csv function which includes past sales trends & seasonal variations

Step 2: Data processing - handle missing data using imputer & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build Regression model using linear regression,SVR, decision tree

Step 6: evaluate the model accuracy using RSquare, SSE or SSR to figure out which model is best

Step 7: Deploy the best model to predict the future electricity consumption

8. An email service provider wants to filter spam emails.

- a) What type of machine learning approach should be used?
- b) Explain the process step by step.

Model type: Classification

Step 1: Collect the email dataset using read_csv function

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: evaluate the model using accuracy, f1, classification matrix, precision to figure which model is best

Step 7: Deploy the best model to predict whether the mail is spam or not

9. A ride-hailing company wants to cluster drivers based on their driving behavior and trip frequency.

- a) Which ML technique should be applied?
- b) Describe the steps for implementation.

Model type - Clustering

Step 1: Collect the drivers dataset using read_csv function which includes driving behavior & trip frequency

Step 2: Data processing - handle missing data & encode categorical data

Step 3: Create a model using clustering methods like DBSCAN, OPTICS, agglomerative clustering, K means etc.

Step 4: Add Hypertuning to the model & using elbow method choose the best model

Step 5: Deploy the best model & start categorizing drivers

10. A medical research institute wants to predict disease progression based on patient health records.

- a) What machine learning approach should be used?
- b) Outline the steps to develop the model.

Model type: Classification

Step 1: Collect the health dataset using read_csv function which includes the patient health records

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: evaluate the model using accuracy, f1, classification matrix, precision to figure which model is best

Step 7: Deploy the best model to predict whether the disease is progressed or not